

Your grade: 100%

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Next item →

1.

Identify the characteristics of convolutional neural networks (CNNs) that make them suitable for image classification.

1 / 1 point
- ☐ Requirement of fixed-size input.

☒ Translation invariance.

✔

Correct

Correct. CNNs exhibit translation invariance, meaning they can recognize objects regardless of their position in the image.

☐ Increased number of parameters compared to fully connected networks.

☒ Local connectivity of neurons.

✔

Correct

Correct. Local connectivity allows CNNs to capture spatial features and hierarchies in images effectively.

☒ Weight sharing in convolutional layers.

✔

Correct

Correct. Weight sharing helps in reducing the number of parameters and capturing spatial hierarchies in the image.
2.

What is the benefit of using the MNIST dataset for training a convolutional GAN?

1 / 1 point
- ☐ It contains colored images, which are ideal for testing

☐ It has a large number of classes, making it suitable for complex models

☐ The images are high-resolution, which enhances model accuracy

☒ It is a well-studied dataset with a simple structure

✔

Correct

Correct! The MNIST dataset's simplicity and extensive use in research make it ideal for training and experimenting with GANs.

3.

Which of the following is the primary purpose of using a convolutional layer in a GAN when working with the MNIST dataset?

1 / 1 point

☐ To convert images to grayscale

☐ To increase the dimensionality of the dataset

☐ To perform data augmentation

☒ To extract spatial features from the images

✔

Correct

Correct! Convolutional layers are designed to extract spatial features from images, which is crucial for tasks involving image data.

4.

Which of the following are key aspects of defining a generator function for a DCGAN?

1 / 1 point

☒ Incorporating batch normalization

✔

Correct

Correct! Batch normalization helps in stabilizing the training process.

☒ Utilizing transposed convolutional layers

✔

Correct

Correct! Transposed convolutional layers are essential for upsampling in the generator.

☐ Adding max pooling layers

☐ Applying L2 regularization to weights

☒ Using a sigmoid activation function in the output layer

✔

Correct

Correct! The sigmoid activation function is often used in the output layer to produce pixel values between 0 and 1.

5.

What is the primary purpose of the DCGAN generator function?

1 / 1 point

☐ To encode images into a latent space

☐ To reduce the dimensionality of image data

☐ To classify images into different categories

☒ To convert random noise into meaningful images

✔

Correct

Correct! The generator function transforms random noise into coherent images.